Jacopo Vitale

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Computer Science Engineer

Profile

PhD Candidate in Bioengineering and Intelligent Systems with a strong focus on **Medical AI**, **Human-Technology Interaction**, and **Health Tech**. My expertise lies in designing full stack Deep Learning architectures for clinical decision support and **Clinical NLP**, grounded in a solid understanding of **Biomedical Signal Processing** and multi-modal data analysis (Echocardiography, MRI, ECG, Omics). During my PhD, I have contributed to high-impact **EU Projects**, gaining valuable experience in collaborative international research. I am currently seeking an R&D position in the industry, aiming to apply my technical skills and innovative mindset to develop cutting-edge healthcare solutions.

Education

Sep 2026 (Expected) **PhD in Bioengineering & Intelligent Systems**, *University Campus Bio-Medico of Rome*, Rome, IT
- **Labs:** Intelligent Health Technology (IHT) Lab & Neurophysiology (NeXT) Lab
- **Focus:** Artificial Intelligence for Healthcare, Medical Image Analysis, Human-Technology Interaction

Dec 2022 **MSc in Computer Science Engineering**, *University of Cassino and Southern Lazio*, Cassino, IT, 110/110 *cum Laude*
- **Thesis:** *Deep Learning strategies for Covid-19 prognosis based on Genomic data*

Sept 2020 **BSc in Computer Science and Telecommunications Engineering**, *University of Cassino and Southern Lazio*, Cassino, IT
- **Thesis:** *Random Vector Functional Link Network porting, from Matlab to C for MCU ESP32*

Research Experience

2022 – Present **Doctoral Researcher**, *University Campus Bio-Medico of Rome*, Rome, IT
Joint Supervision: IHT Lab & Neurophysiology (NeXT) Lab
- **Robotic TMS Automation:** Developed a Bayesian Gaussian Process framework to automatically localize the motor M1 hotspot in a robotic Transcranial Magnetic Stimulation platform.
- **Rare Disease Diagnostics:** Engineered an integrated platform for rare disease classification using multi-modal ophthalmic (fundus) and cardiovascular data.
- **Human-Technology Interaction:** Designed hybrid Genetic Algorithms for simultaneous feature selection and hyperparameter tuning in embodiment experiments.
- **Real-time Pupillometry:** Built a custom CNN-based platform (TypeScript/Vite) for real-time pupil size extraction and experimental monitoring.
- **Oncology & Data Science:** Applied ML to predict Imatinib resistance in GIST tumors and HER2 expression in colorectal cancer.

Aug 2024 – Nov 2024 **Visiting PhD Researcher**, *University Medical Center (UMC) Utrecht*, Utrecht, NL
Research Unit of Utrecht Patient Oriented Database (UPOD)
- **Automatic EHR Classification:** Developed Natural Language Processing and Data Mining pipelines for cardiac risk management in geriatric patients on a dataset of 3,482 patients.

May 2022 – Sep 2022 **Research Intern (AIDA Lab)**, *University of Cassino*, Cassino, IT
- Investigated genomic strategies to classify Covid-19 patient outcomes and implemented feature extraction.

Publications

[1] **Vitale, J.**, et al. "Artificial intelligence for left ventricular hypertrophy detection and differentiation on echocardiography, cardiac magnetic resonance and cardiac computed tomography: A systematic review." *International Journal of Cardiology*, 2025.

[2] **Vitale, J.**, et al. "Automatic Text Classification in Cardiac Risk Management: A Pilot Study." *NBC on BME*, 2025.

[3] **Vitale, J.**, et al. "Rare Eye Diseases Automatic Classification: A Deep Learning Approach." *EMBEC*, 2024.

[4] **Vitale, J.**, et al. "A Machine Learning Approach for Predicting Electrophysiological Responses in Genetically Modified HEK Cells." *EMBEC*, 2024.

[5] Formica, V., . . . , **Vitale, J.**, et al. "Clinicomics for predicting HER2 expression in metastatic colorectal cancer: a multicenter machine learning analysis on real-world data." *Oncology*, 2025.

[6] Sassi, M., **Vitale, J.**, et al. "Genetic Algorithm for Predicting Primary Imatinib Resistance in Locally Advanced or Metastatic Gastrointestinal Stromal Tumors." *NBC on BME*, 2025. (**Best Research Paper Award**)

[7] Pagnano, M.E., . . . , **Vitale, J.**, et al. "Echocardiographic Image-Based Classification of Cardiac Amyloidosis: A Proof of Concept." *NBC on BME*, 2025.

Software & Tools Developed

- **eHTA Tool:** Online automated platform for Early Health Technology Assessment.
- **Rare Path Platform:** AI diagnostic platform for rare disease classification.
- **Pupil Extraction:** Real-time web-based pupillometry tool (TypeScript/Vite, CNN).
- **TMS Hotspot:** ML-driven robotic localization tool using Bayesian Gaussian Processes.

Technical Skills

AI & GUI Dev	Python (PyTorch, Pandas, GUI Frameworks), MATLAB (App Designer), Deep Learning, NLP.
Web Development	TypeScript, Node.js, Vite, Real-time Web Applications.
Embedded & 3D	Raspberry Pi, Microcontrollers (ESP32), 3D Design (Fusion 360).
DevOps & Systems	Docker, Kubernetes, Linux Administration, Bash Scripting, Git.
Biomedical	Medical Image Analysis (DICOM), Signal Processing (ECG, HRV, EEG).

Awards & Service

- **Best Research Paper Award** – 20th NBC & 24th PCBBE, Warsaw (2025).
- **Winner, AI Application in Healthcare** – AIIC 2024, Rome.
- **Winner, Scientific Challenge** – EMBEC 2024, Slovenia.
- **Organising Committee Member** – IEEE BHI'24 Data Challenge.
- **Guest Lecturer** – University of Cambridge (2024). Seminar on Docker & Kubernetes.
- **Peer Reviewer** – Scientific Reports, IEEE Sensors Journal.

Languages

Italian	Native Speaker
English	Full Professional Proficiency

Additional Information

Driving License	Category B (Italian/EU)

I hereby authorize the processing of the personal data contained in this CV in compliance with the GDPR (EU Regulation 2016/679).

Last updated: January 12, 2026